SC7 Bayes Methods MT26

Lecturer: Geoff Nicholls

Lecture 5: The Savage Axioms and subjective preference

Probability distributions - notation reference slide

Let S be a sample space and let $\mathcal{S}$ be a σ-field of sets in S satisfying $S \in \mathcal{S}$, $A \in \mathcal{S} \Rightarrow A^c \in \mathcal{S}$ and if

$$A_1, A_2, ... \in \mathcal{S} \ \Rightarrow \ \bigcup_{i=1}^{\infty} A_i \in \mathcal{S}.$$

For example if S is finite then $\mathcal{S}$ can be taken as the power set. A probability distribution on $(S, \mathcal{S})$ is (i) a map $P : \mathcal{S} \to [0, 1]$ satisfying (ii) $P(S) = 1$, and if $A_1, A_2, ... \in \mathcal{S}$ are disjoint then

$$P(\cup_i A_i) = \sum_i P(A_i) \qquad \text{(iii) "countably additive"}$$

The triple $(S, \mathcal{S}, P)$ is called a probability space.

The Savage Axioms (following DeGroot*)

If prior preferences satisfy axioms 1-5 $\Rightarrow$ prior exists.

If preferences over $P_\delta,\ \delta \in \mathcal{D}$ satisfy axioms 6-10 $\Rightarrow$ utility exists.

We begin with the Savage axioms for qualitative probability.

Theorem: *Let S be a sample space and let $\mathcal{S}$ be a σ-field of sets in S. Let a system $\succ, \prec, \sim$ of preferences over $A, B \in \mathcal{S}$ be given. A probability distribution $\pi(A), A \in \mathcal{S}$ expressing these preferences exists and is unique if and only if the preference relations satisfy Savage Axioms 1-5.*

If a system of preferences $\succ, \prec, \sim$ over $A, B \in \mathcal{S}$ satisfies the Savage axioms then a unique probability distribution expressing those preferences exists, so $\pi(A) > \pi(B) \Leftrightarrow A \succ B$ etc.

*in "Optimal Statistical Decisions" (1970), see also CR-TBC

Axioms of Preference

Axiom 1. For any two events A and B exactly one of the following relations must hold: $A \succ B$, $A \prec B$, $A \sim B$.

Axiom 2. If $A_1 \cap A_2 = B_1 \cap B_2 = \emptyset$ and $A_i \succeq B_i, i = 1, 2$ then

$$A_1 \cup A_2 \succeq B_1 \cup B_2.$$

If in addition either $A_1 \succ B_1$ or $A_2 \succ B_2$ then $A_1 \cup A_2 \succ B_1 \cup B_2$.

If A can happen two exclusive ways, and each is more likely than corresponding events leading to B then A is more likely then B.

Axiom 3. If $A \in \mathcal{S}$ then $\emptyset \preceq A$. Furthermore $\emptyset \prec S$.[†]

It follows (PS2) from Axioms 1-3 that the order is transitive (if $A \preceq B$ and $B \preceq C$ then $A \preceq C$) and $A \preceq B \Rightarrow A^c \succeq B^c$.

[†]to avoid trivial case where all sets $A \sim \emptyset$.

Fixed odds-preference in the Ellsberg paradox doesn't satisfy Axiom 2. If $S = \{(r,r),(r,b),(b,r),(b,b))\}$ is the set of outcomes when a ball is drawn from each urn, then

$$\begin{aligned} r_A &= \{(r,r),(r,b)\}, \quad r_B = \{(r,r),(b,r)\}, \\ b_A &= \{(b,r),(b,b)\}, \quad b_B = \{(r,b),(b,b)\}. \end{aligned}$$

Identify

$$A_1 = r_A, \quad A_2 = b_A, \quad B_1 = b_B, \quad \text{and} \quad B_2 = r_B.$$

The elicited preferences included $A_1 \succ B_1$ and $A_2 \succ B_2$. By Axiom 2 we have

$$A_1 \cup A_2 \succ B_1 \cup B_2,$$

but $A_2 = A_1^c$ and $B_2 = B_1^c$ so $S \succ S$, a contradiction.

Axioms 4 and 5 and the existence of π[‡]

Axiom 4 ensures our final $\pi(A)$ will be countably additive.

Axiom 4: If $A_1 \supset A_2 \supset \ldots$ is a decreasing sequence of events with $A_i \succeq B$ for each $i = 1, 2, \ldots$ and fixed B then $\bigcap_{i=1}^{\infty} A_i \succeq B$.

Axiom 5 assigns probabilities. Let $X \sim U(0, 1)$.

Replace S with $S' = S \times [0, 1]$ and $\mathcal{S}$ with $\mathcal{S}' = \mathcal{S} \otimes \mathcal{B}_{[0,1]}$. Require A1-A4 for sets in $\mathcal{S}'$.

Axiom 5: for all $p \in [0, 1]$, $\{X \leq p\} \in \mathcal{S}'$.

We can now determine $\pi(A), A \in \mathcal{S}$ (old $\mathcal{S}$). We prove there is p so that $A \sim \{X \leq p\}$ and set $\pi(A) = p$.

The probability space $(S, \mathcal{S}, \pi)$ defined in this way will satisfy the axioms of probability.

[‡]the material on this slide is outside the syllabus

Relation to Axioms of probability (AoP)

When the Savage Axioms hold, any statement about preferences can be verified as a statement about probability.

Example: for $A, B \in \mathcal{S}$ suppose $A \subseteq B$. We must have $A \preceq B$ from both sets of axioms.

From AoP (assume π expressing preference exists)
$\pi(B) = \pi(A) + \pi(B \setminus A)$ by AoP(iii) so $\pi(A) \leq \pi(B)$ by AoP(i).
But π expresses preferences so $A \preceq B$.

From SA's (without reference to π)
Claim (PS2): If $A_1 \cap D = B_1 \cap D = \emptyset$ then $A_1 \cup D \succ B_1 \cup D$ iff $A_1 \succ B_1$ by A1-3.

If $A_1 = \emptyset$, $D = A$ and $B_1 = B \setminus A$ then $A \succ B$ is $A_1 \cup D \succ B_1 \cup D$ implying $\emptyset \succ B \setminus A$ by claim, contradicting Axiom 3.

Axioms of utility

Let $\mathcal{P}_D = (P_\delta)_{\delta \in \mathcal{D}}$ be all reward distributions. For $P \in \mathcal{P}_D$,

$$\Pr(r_{\min} \leq R \leq r_{\max}) = 1.$$

Theorem: *There exists a utility function U which expresses our preference relations over $P \in \mathcal{P}_D$ if and only if our preferences over reward distributions satisfy Savage Axioms 6-10.*

If we give preference relations $\succ_U$, $\prec_U$ and $\sim_U$ over reward distributions $P \in \mathcal{P}_D$ satisfying Savage Axioms 6-10 then there exists a utility function U expressing those relations from its expected utility, so that

$$E_P(U(r)) > E_{P'}(U(r)) \quad \Leftrightarrow \quad P \succ_U P'.$$

The first two axioms state that the order relates all lotteries $P, P' \in \mathcal{P}_D$ and is transitive.

Axiom 6: For any two reward distributions $P, P' \in \mathcal{P}_D$ exactly one relation must hold: $P \succ_U P'$ or $P \prec_U P'$ or $P \sim_U P'$.

Axiom 7: If $P \succeq_U P'$ and $P' \succeq_U P''$ then $P \succeq_U P''$.

The next axiom says that the preference between two rewards (or distributions over rewards) should not change if we alter both in the same way.

Axiom 8: let $0 < \alpha < 1$ and $P, P', P'' \in \mathcal{P}_D$ be given. Then $P' \succ_U P''$ if and only if

$$\alpha P' + (1-\alpha)P \succ_U \alpha P'' + (1-\alpha)P.$$

The choices we made in the Allais paradox violate Axiom 8. If

$$P' = (0, 1, 0), \quad P'' = (1/11, 0, 10/11)$$

$$P = (0, 1, 0), \quad \tilde{P} = (1, 0, 0)$$

then the reward distributions for lotteries A-D can be written

$$\begin{array}{rcll} p^{(A)} & = & (0, 1, 0) & = 0.11P' + 0.89P \\ p^{(B)} & = & (0.01, 0.89, 0.1) & = 0.11P'' + 0.89P \\ p^{(C)} & = & (0.89, 0.11, 0) & = 0.11P' + 0.89\tilde{P} \\ p^{(D)} & = & (0.9, 0, 0.1) & = 0.11P'' + 0.89\tilde{P} \end{array}$$

Since P is the same for options (A) and (B) and we prefer (A) we have $P' \succ_U P''$ by Axiom 8. Similarly, $\tilde{P}$ is the same for (C) and (D) so a preference for (D) leads to $P' \prec_U P''$, a contradiction.

Axioms 9 and 10 and the utility existence proof

I omit Axioms 9 and 10. See CR-TBC/deGroot for detail. The lecture notes include some discussion included for completeness but *outside the scope of the course.*

Conclusions: The Savage axioms are prescriptive - if our preferences do not satisfy the axioms, then they should be altered.

If they dont hold then we cant do coherent inference. Do the paradoxes lead us to dismiss this notion of coherence? Allais in particular has motivated research on ideas of utility.

The SA are a careful answer to a question we dont always have to ask. If the prior is based on a stochastic model (as in Lecture 2) it exists and the choice of loss function is often obvious.

For a more interesting question try "is this prior representative of available information?".

de Finetti's Theorem gives sufficient conditions for a "true prior" (not just expressing subjective preferences) to exist. But we may satisfy the SA with a bad prior. Hence Prof David Cox's criticism of the Savage Axioms: "what is missing is the truth".